

RAJESH VHANKADE

AI/ML Engineer | Generative AI | RAG | LangChain | FastAPI | Python | AWS

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Pune, Maharashtra, India

PROFESSIONAL SUMMARY

AI/ML Engineer with 2+ years of experience building Generative AI applications, Retrieval-Augmented Generation (RAG) systems, and backend services using Python and FastAPI. Experienced in developing document search and question-answering solutions using LLMs, vector databases, and retrieval pipelines. Hands-on experience with LangChain, MCP, OpenAI APIs, AWS, and workflow automation. Focused on building practical AI solutions that improve information access and business processes.

TECHNICAL SKILLS

- **Generative AI & LLMs:** RAG, LangChain, LangGraph, MCP (Model Context Protocol), Prompt Engineering, OpenAI APIs, Hugging Face Transformers
- **AI/ML:** NLP, Embeddings, Semantic Search, Similarity Search, Text Classification
- **Retrieval & Search:** FAISS, ChromaDB, Hybrid Search, Metadata Filtering, Re-ranking
- **Backend Development:** Python, FastAPI, REST APIs
- **Databases:** PostgreSQL, MongoDB, SQLite
- **Cloud & DevOps:** AWS, Docker, CI/CD
- **Tools:** Git, GitHub, Postman

PROFESSIONAL EXPERIENCE

SmartMatrix Digital Services Pvt Ltd. | April 2024 - Present

AI/ML Engineer

- Built and maintained Generative AI applications for enterprise use cases using LLMs, retrieval pipelines, and FastAPI services.
- Developed APIs and backend components supporting document search, question answering, and AI-assisted workflows.
- Implemented retrieval pipelines using embeddings, vector databases, metadata filtering, and hybrid search techniques.
- Integrated OpenAI and Hugging Face models into business applications through prompt-based workflows.
- Worked on document ingestion, preprocessing, chunking, embedding generation, retrieval, and response generation pipelines.
- Improved application responsiveness through asynchronous processing and optimized retrieval logic.
- Containerized services using Docker and deployed applications on AWS infrastructure.
- Collaborated with business and technical teams to convert requirements into deployable AI solutions.

Radixle Technology Solutions | Oct 2023 – Mar 2024

Machine Learning Engineer Intern

- Assisted in developing machine learning solutions through data preprocessing, feature engineering, model training, validation, and performance evaluation using Python.
- Supported data-driven product initiatives by collaborating with development teams to integrate ML functionality into business applications.
- Contributed to model experimentation and optimization efforts to improve prediction quality and system performance.

PROJECT EXPERIENCE

Enterprise Knowledge Assistant

Domain: Manufacturing Operations & Knowledge Management

- Developed an AI-powered assistant to help manufacturing teams search and access information from SOPs, maintenance manuals, troubleshooting guides, and internal documentation.
- Built a Retrieval-Augmented Generation (RAG) pipeline covering document ingestion, text chunking, embedding generation, vector indexing, retrieval, and answer generation.
- Implemented semantic search and metadata-based filtering to improve retrieval relevance for technical documents.
- Designed prompts and response templates to generate answers grounded in retrieved content.
- Developed FastAPI endpoints for document search and question-answering functionality.
- Used asynchronous processing and caching to improve API response times.
- Deployed the application using Docker and AWS EC2.
- Tech Stack: Python, FastAPI, OpenAI API, LangChain, FAISS, PostgreSQL, Docker, AWS EC2

HR-Payroll Data Synchronization Platform

Domain: HR & Payroll Process Automation

- Developed a prototype system to synchronize employee data changes between HR and Payroll databases using MCP (Model Context Protocol).
- Created MCP servers exposing tools for detecting employee record changes, validating updates, generating sync payloads, and applying changes to the target system.
- Implemented SQLite triggers to automatically capture employee updates and store change records for processing.
- Built workflow logic to validate employee changes before transferring data between systems.
- Generated structured JSON payloads containing employee updates and synchronization details.
- Applied approved updates to the payroll database to keep employee information consistent across systems.
- Simulated a common business scenario where salary and employee profile updates made in HR systems are reflected in Payroll systems.
- Tech Stack: Python, LangChain, MCP (Model Context Protocol), SQLite, JSON, Git

PROFESSIONAL DEVELOPMENT

Data Science & Machine Learning Program | Yess Infotech, Pune | Sep 2022 - Sep 2023

- Completed training in Python, Statistics, Machine Learning, Deep Learning, NLP, and Data Science.
- Built projects involving data preprocessing, model training, classification algorithms, and evaluation techniques.
- Gained practical experience with machine learning workflows and deployment concepts.

EDUCATION

- **Bachelor of Engineering (Computer Engineering)**
Savitribai Phule Pune University | 2022 | 8.96 CGPA
- **Diploma in Computer Engineering**
Government Polytechnic, Miraj | 2019 | 71.96%